DESIGN AND DEVELOPMENT OF A HOPPER AND CONVEYOR SYSTEM FOR AUTOMATED TiO_2 FILLING, WEIGHING, AND SEALING

A Mini Project Report

Submitted to the APJ Abdul Kalam Technological University

in partial fulfillment of requirements for the award of degree

Master of Technology

in

ELECTRONICS and COMMUNICATION

with specialisation

Robotics and Automation

by

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MINI PROJECT

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

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2024-2026

CERTIFICATE

This is to certify that the report entitled **DESIGN AND DEVELOPMENT OF A HOPPER AND CONVEYOR SYSTEM FOR AUTOMATED TiO_2 FILLING, WEIGHING, AND SEALING** submitted by **SHARANSUNDAR.S** (TVE24ECRA13) to the APJ Abdul Kalam Technological University in partial fulfillment of the M.Tech. degree in Robotics and Automation is a bonafide record of the Mini Project work carried out by him under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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DECLARATION

I, the undersigned, hereby declare that the project report submitted in partial fulfillment of the requirements for the award of the degree of Master of Technology of the APJ Abdul Kalam Technological University, Kerala, is a bonafide work done by me. This submission represents the ideas in my own words, and where ideas or words of others have been included, they have been adequately and accurately cited and referenced to the original sources.

I also declare that I have adhered to the ethics of academic honesty and integrity and have not misrepresented or fabricated any data, idea, fact, or source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have not been properly cited or from whom proper permission has not been obtained.

This report has not been previously formed as the basis for the award of any degree, diploma, or similar title of any other University.

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July, 2025

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ABSTRACT

This project presents the design and development of an automated prototype intended to improve the weighing, filling, and sealing processes for Titanium Dioxide (TiO_2) at Travancore Titanium Products Limited. The existing manual methods used at the company expose workers to health risks due to direct handling of the chemical and result in inconsistencies in the packaging process. To address these challenges, a scaled-down prototype has been developed that demonstrates the potential for automation through the integration of key subsystems including a hopper with a servo-controlled butterfly valve, a load cell-based weighing platform with tilting functionality, a heat sealing mechanism, and an automated conveyor system. Arduino Mega serves as the central control unit, managing these subsystems in sequence based on sensor feedback.The specific focus of this report is on the design and development of the hopper system, the conveyor system, and their integration within the overall prototype. The hopper was modeled as a modular assembly to control material flow precisely, while the conveyor system was developed to automate the transfer of packets through IR-sensor-triggered motion and stepper motor control. Though this prototype handles only 100-gram packets as a proof of concept, it sets a strong foundation for future industrial-scale adaptations. Proposed future enhancements include scaling for 25-kilogram capacity, implementing ultrasonic sealing for tamper-proof packaging, and integrating automated packet feeding. This work illustrates how cost-effective automation strategies can significantly improve both the safety and efficiency of material handling operations in industrial environments.

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Chapter 1

Introduction

1.1 Background

Titanium Dioxide (TiO_2) is a highly versatile industrial chemical, commonly used in paints, coatings, plastics, and cosmetics. Its widespread use is largely attributed to its excellent opacity and brightness properties. At Travancore Titanium Products Limited (TTPL), Titanium Dioxide is produced in large quantities and requires efficient, safe, and accurate packaging systems. However, the current packaging process at TTPL involves significant manual intervention, which presents several challenges.

Manual handling of TiO_2 poses serious health risks to the workforce. Prolonged exposure to fine TiO_2 powder without adequate safety measures can lead to respiratory issues, skin irritation, and other health hazards. In addition to the health risks, manual filling processes are prone to inconsistencies in packet weight and are inefficient in terms of labor and time utilization. Figure 1.1 shows a photograph of the current manual filling system employed at TTPL.

Recognizing these limitations, this project was initiated to develop an automated prototype system that can simulate an industry-relevant filling, weighing, sealing, and conveying process for Titanium Dioxide. The goal of the prototype is to establish a foundation for transitioning towards safer, more consistent, solutions.

Figure 1.1: Existing Manual Filling System at Travancore Titanium Products Limited (TTPL)

1.2 Problem Statement

The current manual system at TTPL exposes operators to health hazards and suffers from inefficiencies and inconsistencies in packet filling. A lack of automation in these processes leads to variable packet weights, increased material wastage, and inconsistent sealing quality, ultimately affecting productivity and quality control.

The fundamental issue is the absence of an integrated system that can:

- Accurately measure the material weight during filling.
- Control the flow of TiO_2 efficiently and precisely.
- Provide consistent, tamper-proof sealing.
- Transfer packets automatically without manual handling.

1.3 Objectives

The primary aim of this project is to develop a scaled-down prototype capable of automating the critical processes of weighing, filling, sealing, and transferring TiO_2 packets. The specific objectives are as follows:

- Design and develop a modular hopper system with controlled powder flow.
- Build a conveyor mechanism to automate the packet movement.
- Integrate a tilting mechanism for packet discharge post-weighing.
- Implement a sealing mechanism to ensure packet integrity.
- Synchronize all subsystems through a centralized control unit (Arduino Mega).

This report emphasizes the work completed on the hopper system, the conveyor system, and their seamless integration with the overall prototype, aligning with the focused contributions of the project team.

1.4 Scope of the Work

The prototype is developed to handle packets of up to 100 grams, primarily for demonstration and validation of automation feasibility. It integrates key components including:

- A load cell for precise weight measurement.
- A servo-controlled butterfly valve for controlled powder flow.
- A heat sealing mechanism actuated via servo motor.
- An IR sensor-triggered conveyor system powered by a stepper motor.
- An Arduino Mega for centralized control of operations.

Although this model is limited to a small scale, it serves as a proof-of-concept for future industrial upscaling. Future iterations are proposed to handle larger weights (up to 25 kg), replace the heat sealing with ultrasonic sealing for tamper-proofing, and introduce automated packet feeding systems. These upgrades aim to fully align the system with the needs and expectations outlined by TTPL through their provided questionnaire.

Chapter 2

Literature Survey and Company Requirements

2.1 Introduction

The growing demand for process automation in industrial sectors is driven by the need to enhance precision, improve productivity, ensure workplace safety, and minimize human errors. Industries involved in the packaging of chemical powders, such as Titanium Dioxide (TiO_2), are increasingly recognizing the benefits of automation in ensuring consistent quality and reducing occupational hazards. A detailed review of existing technologies, mechanisms, and industrial practices provides valuable insights for developing an optimized, automated solution tailored to specific operational environments such as that of Travancore Titanium Products Limited (TTPL).

Through this literature survey, relevant studies on material handling, load cell integration, servo-driven actuation, sealing technologies, and conveyor-based transfer mechanisms were analyzed. Previous works highlight how automation has successfully replaced manual intervention in various sectors, leading to improvements in accuracy, repeatability, and safety. The survey also draws attention to the growing adoption of microcontroller-based systems like Arduino for controlling industrial prototypes, offering cost-effective solutions for real-time process management.

In parallel, the company's requirements were studied through discussions with the technical team at TTPL and through a formal questionnaire. The current manual method of filling and packaging TiO involves several health risks, operational inefficiencies, and a lack of tamper-proof sealing, all of which TTPL aims to address through this automation initiative. The company emphasized the need for an automated system capable of accurate weighing, controlled filling, effective sealing, and safe transfer of packets—all while being scalable to handle industrial-scale 25 kg bags in the future.

This chapter thus sets the foundation for understanding the technological landscape, identifies gaps in the current manual processes, and defines the specific requirements that guided the design and development of the proposed prototype.

2.2 Literature Survey

2.2.1 Health Risks Associated with Manual Handling of Titanium Dioxide

Studies and industry reports consistently highlight the health hazards associated with prolonged exposure to fine chemical powders like Titanium Dioxide. Inhalation of airborne TiO_2 particles, especially during bagging and transfer processes, has been linked to respiratory complications, skin irritation, and long-term occupational health risks. As per industrial safety guidelines, automated systems with minimal human intervention are recommended to mitigate such exposures.

2.2.2 Need for Accurate and Consistent Weighing

Literature on automated packaging solutions emphasizes the importance of integrating precise weighing mechanisms into bulk material handling processes. Load cells, when coupled with microcontrollers, enable real-time feedback and automated filling control, ensuring consistency and reducing material wastage. This aligns with modern industry expectations for maintaining product uniformity.

2.2.3 Advances in Sealing and Conveying Systems

Automation advancements have also impacted sealing and conveying technologies. Ultrasonic sealing has been widely adopted for its reliability and tamper-proof qualities in various industries, although heat sealing remains a viable option for prototypes. Automated conveyors driven by sensor feedback and controlled through microcontrollers enhance efficiency by streamlining the transfer of filled packets.

2.3 Current Practices at TTPL

Currently, the filling, weighing, and sealing of Titanium Dioxide at TTPL is conducted manually. This exposes workers directly to TiO_2, increasing health risks, and introduces inconsistencies in packet weight and sealing quality. Moreover, manual handling lacks the efficiency and repeatability required for modern manufacturing standards.

Figure 2.1: Manual Filling , sealing System in Operation at TTPL

2.4 Company Requirements (Based on Questionnaire)

2.4.1 Key Requirements Identified

- Complete automation of weighing, filling, and sealing processes.
- Elimination of direct human contact with TiO_2 powder.
- Reliable weight accuracy for packets up to 25kg.
- Implementation of tamper-proof sealing, with ultrasonic sealing preferred.
- Inclusion of a conveyor mechanism for automated packet transfer.
- Robust system suitable for dusty industrial environments.

2.4.2 Filled Questionnaire from TTPL

The filled questionnaire provided by TTPL forms the basis for defining the objectives and future proposals of this project. It outlines both current challenges and the desired features of a future solution.

Bag Filling Questionnaire

Company Name	M/s. Travancore titanium product Ltd
Address	Kochuveli ,thiruvananthapuram,695021
Contact Person	Dilip p (manager instrumentation), shabin m n ,Dy mgr packing
Contact Number	Dilip 8086397933, Shabin 9495063810
Installation Location/ Site Address	TTPL ,thiruvananthapuram
General	
Equipment -	Bag Filling System
Type- semi auto bag filling and stitching	Open Mouth –yes
	Currently open mouth bagging hdpe bas
	Will explore options for valve bags and paper bags
Product Specifications	
Product to be bagged	Titanium dioxide powder(anatasegrade, rutile grade)
Bagging Temperature	35-40^{0}C
Bulk Density	Anatase 1 gm/cm3, Rutile 0.8gm/cm3
Moisture content	Anatase0.5% Rutile 0.5%
Flow ability	Highly cohesive,fluffy, not free flowing tendency to stuck
Grain / Particle Size	.3-.45micrometer
Corrosive/Abrasive	Moderately abrasive,non corrosive
Is the material Hygroscopic	No ,moisture ingress must be prohibited
MOC required	MS for normal parts ,SS recommended for portions that directly feed material to bags
Bagging Specifications	
Bag capacity/Capacities	Currently using 25 kg bag
Accuracy	+-5 gms
Bagging Speed	5-7 bags per minute
Overall Height (Available)	Bag height 108cm including liner
Bag Dimensions	90*57cm(108cm length including liner)
Storage Silo Capacity (If existing)	8MT
Existing Arrangement/Feeder at Site	Manual bagging and stiching
From where is the material being fed to the bagging machine	Discharge bunker of pulvariser mill
Do you require a intermediate storage	Not necessary
(Please mail us photos of the existing arrangement if any)	
Type of Bag (HDPE/Paper). Inner Liner if any.	HDPE with inner liner
Any Other specifications:	
Sealing/ stitching machine & conveyor required	Sealing/stitching required
Voltage & Frequency	
Please specify any other requirements.	Please provide a budgetary offer

Figure 2.2: Questionnaire Response from Travancore Titanium Products Limited

2.5 Summary

The literature review highlights the necessity of automation in powder handling for health, safety, and consistency. The current manual methods at TTPL are outdated and expose both the workers and the process to unnecessary risks and inefficiencies. The company's questionnaire provides a clear directive for future developments, guiding this project towards a scalable, industry-relevant solution.

Chapter 3

Hopper System Design and Development

3.1 Introduction

In any powder-based packaging system, the hopper plays a central role in facilitating a controlled and continuous material flow to the filling station. The unique physical properties of Titanium Dioxide (TiO_2) — being fine, lightweight, and prone to clumping — require careful attention in hopper design to avoid material bridging and ensure consistency. For this project, a compact hopper system was designed with the primary objective of delivering precise, consistent quantities of TiO_2 powder during the automated weighing and filling process. This chapter explains the systematic approach followed in designing, fabricating, and integrating the hopper, butterfly valve, and funnel into the working prototype.

3.2 Design Considerations

The hopper system was designed with multiple engineering considerations in mind:

- The material should flow freely under gravity.

- Flow control must be precise to allow accurate weight-based filling.
- The system should be easy to fabricate using cost-effective materials suited for a prototype.
- Future scalability to industrial capacities must remain possible.
- Integration with sensors and actuators should be seamless to support automation.

Considering these factors, the hopper was developed as a modular structure that simplifies maintenance, replacement, and future upscaling.

3.3 Structural Design and CAD Modeling

Using Autodesk Fusion 360, the hopper system was designed as a three-part assembly: the hopper itself, a butterfly valve for flow control, and a funnel with a connector to direct the flow onto the weighing platform. The design prioritizes a gravity-assisted, clog-free flow path from storage to filling point.

3.3.1 Hopper Geometry

The hopper was modeled with a conical profile, tapering towards the butterfly valve opening. This geometry minimizes stagnant zones and material bridging, common issues with fine powders. The walls were given a steep enough angle to ensure self-cleaning flow characteristics, supported further by the valve actuation for periodic opening and closing.

3.3.2 Funnel and Connector Design

The funnel serves to guide the powder in a controlled manner towards the weighing platform. A securely fastened connector ensures the funnel is properly aligned with the valve, preventing leaks and misalignment during operation.

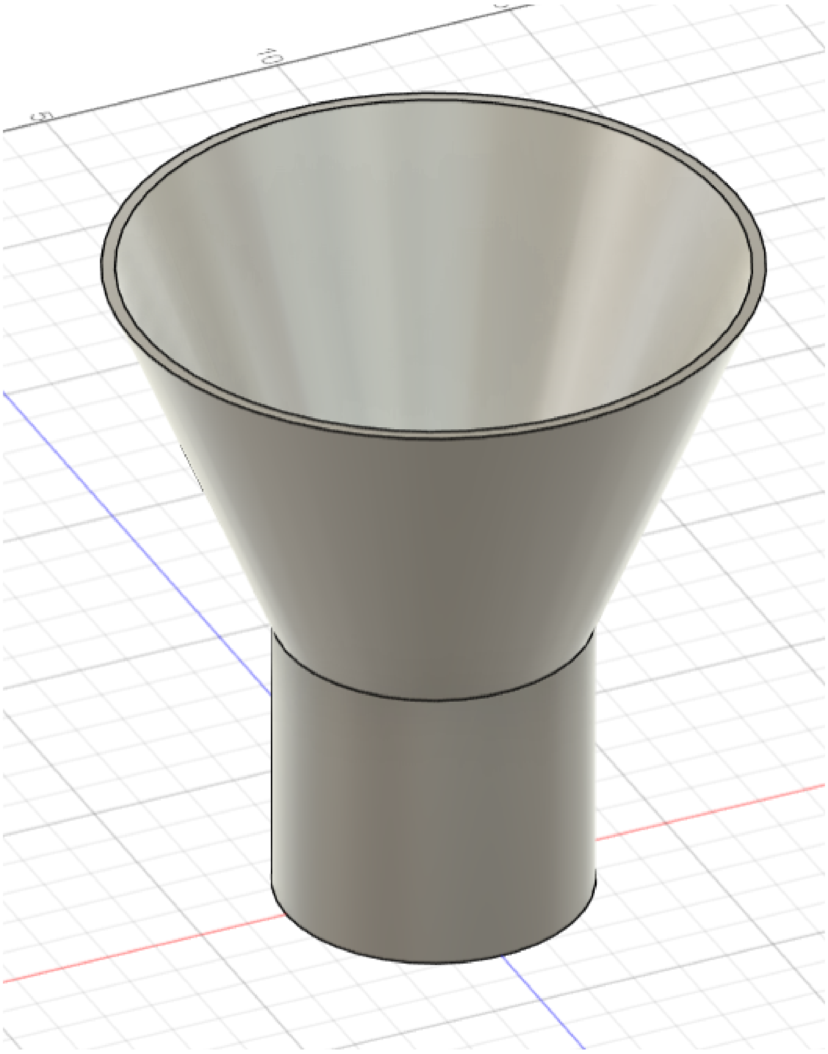

Figure 3.1: 3D Model of Hopper in Fusion 360

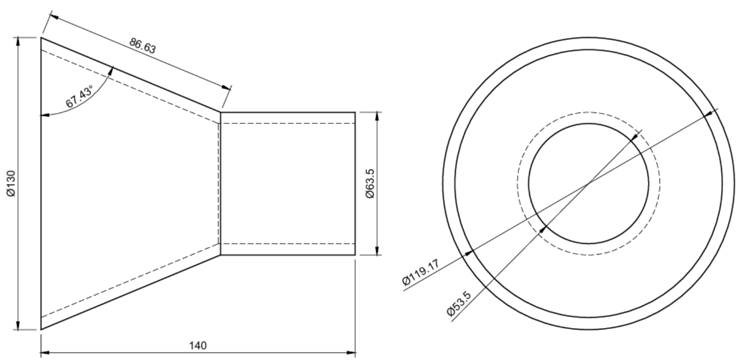

Figure 3.2: Sketch of Hopper Design

3.3.3 Butterfly Valve Mechanism

A servo-actuated butterfly valve was integrated directly beneath the hopper outlet to control the start and stop of the powder flow. This valve operates between 0 degree

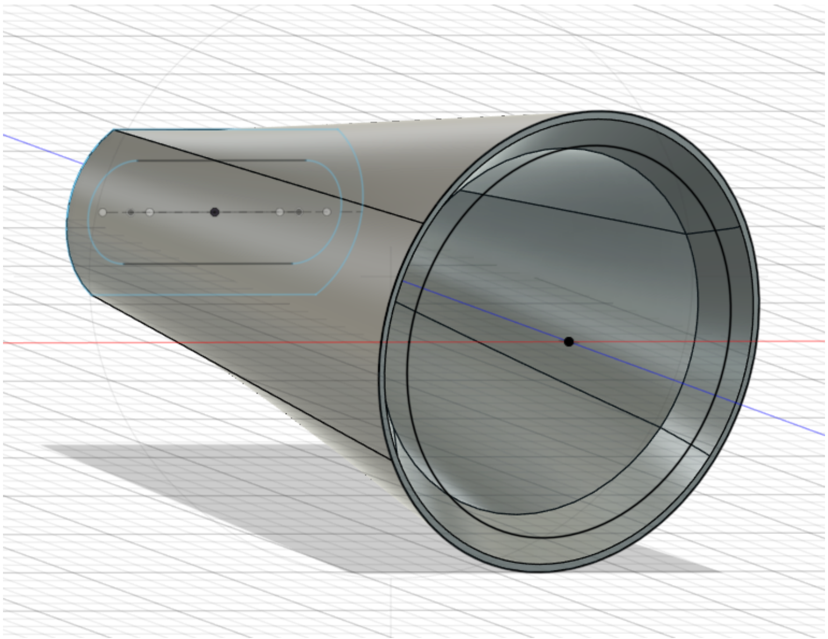

Figure 3.3: 3D Model of Funnel and Connector

(closed) and 90 degree (open) positions, governed by feedback from the load cell platform to achieve precise material dispensing.

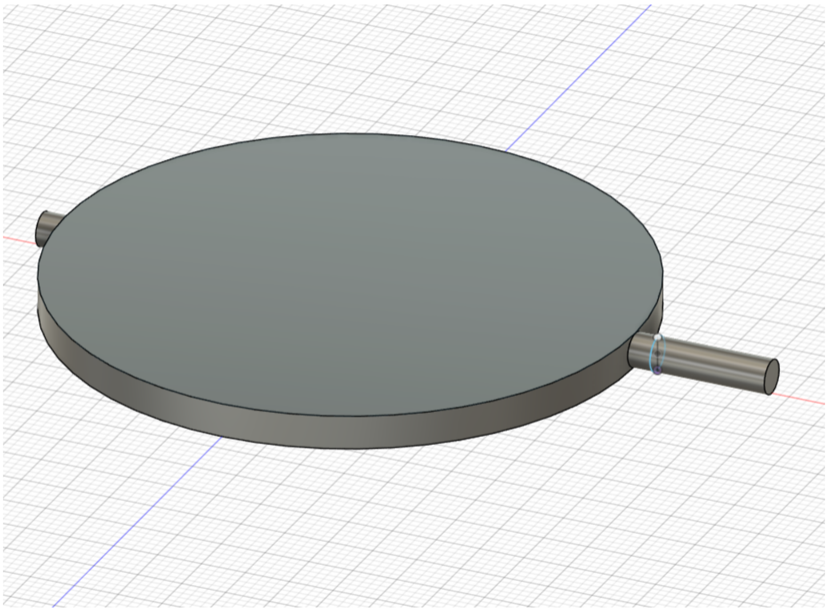

Figure 3.4: 3D Model of Butterfly Valve Assembly

3.4 Fabrication Process

For the prototype stage, the hopper and its components were fabricated using acrylic sheets due to their ease of machining, transparency (for observation during testing), and suitability for lightweight structures. Components were assembled using fasteners to allow disassembly for adjustments or maintenance.

Figure 3.5: Photograph of Assembled Hopper System

3.5 Working Principle

The operation of the hopper system is synchronized with the weighing platform and the butterfly valve. Initially, the servo maintains the valve in the open position (90 degree), allowing powder to flow through gravity onto the bag placed on the load cell platform. As the weight approaches the target (90-100 grams), real-time feedback from the HX711 amplifier prompts the Arduino Mega to rotate the servo, closing the valve (0 degree position) to stop the flow.

This ensures precision in the filling process, reducing material wastage and eliminating overfilling risks. The funnel's geometry guarantees that all dispensed material reaches the intended location without spillage.

3.6 Control System Integration

The butterfly valve's servo motor is directly interfaced with the Arduino Mega controller, receiving control signals based on weight data. The load cell's real-time feedback through the HX711 amplifier forms a closed-loop system, enhancing the accuracy of the filling process.

3.6.1 Arduino Code Snippet

```
#include <Servo.h>
Servo valveServo;
valveServo.attach(9);

if (weight >= 90) {
    valveServo.write(0);  // Close valve to stop filling
}
else {
    valveServo.write(90); // Keep valve open to allow filling
}
```

3.7 Challenges Encountered

Several practical challenges arose during the development of the hopper system:

- Fine TiO_2 powder exhibited static buildup tendencies, affecting smooth flow.
- Ensuring leak-proof connections between modular components required precision in assembly.
- Maintaining consistent servo operation under repeated cycles demanded careful calibration.
- Preventing clogging within the funnel during extended operation.

These challenges were addressed through iterative adjustments in geometry, refining connector fits, and optimizing servo parameters.

3.8 Summary

The hopper system was successfully designed, fabricated, and tested within the scope of this prototype project. It delivers a controlled, accurate flow of material to the weighing platform, effectively integrating mechanical design and electronic feedback control. This system forms a critical part of the overall automation solution, paving the way for scalability and enhancements in future industrial adaptations.

Chapter 4

Conveyor System Design and Development

4.1 Introduction

In any automated material handling setup, the conveyor system plays a crucial role in transferring packets or components smoothly between different stations within the workflow. For this project, the conveyor system was designed specifically to automate the transfer of filled and sealed Titanium Dioxide (TiO_2) packets from the weighing and sealing station to the next point of handling. This chapter provides a detailed account of the conveyor system's design, development, and integration within the overall prototype.

4.2 Design Considerations

The conveyor system was designed to meet the following functional requirements:

- Reliable and smooth transfer of packets without manual intervention.
- Controlled activation based on sensor feedback.

- Precise start-stop mechanism to avoid overshooting or misalignment.
- Lightweight construction suitable for a prototype.
- Easy adaptability for potential future scaling.

Given these considerations, the system was constructed using aluminum extrusion profiles to provide both structural integrity and flexibility during assembly and testing.

4.3 Structural Design and Components

The conveyor structure was built using standard aluminum extrusion profiles for their modularity, ease of assembly, and professional appearance in prototypes. This also allows for future modifications with minimal effort. The conveyor bed was aligned to ensure smooth packet movement without misalignment or jamming.

4.3.1 Mechanical Components

Key components integrated into the conveyor system include:

- Aluminum extrusion profiles for the frame.
- IR sensor for packet detection at the start of the conveyor.
- Limit switch at the end of the conveyor for precise stopping.
- Stepper motor for driving the conveyor belt.
- Belt mechanism to translate motor rotation to linear motion.

4.4 Sensor and Actuation Integration

The conveyor system operates based on input from an IR sensor positioned at the starting point. This sensor detects the presence of a packet as it drops from the tilting

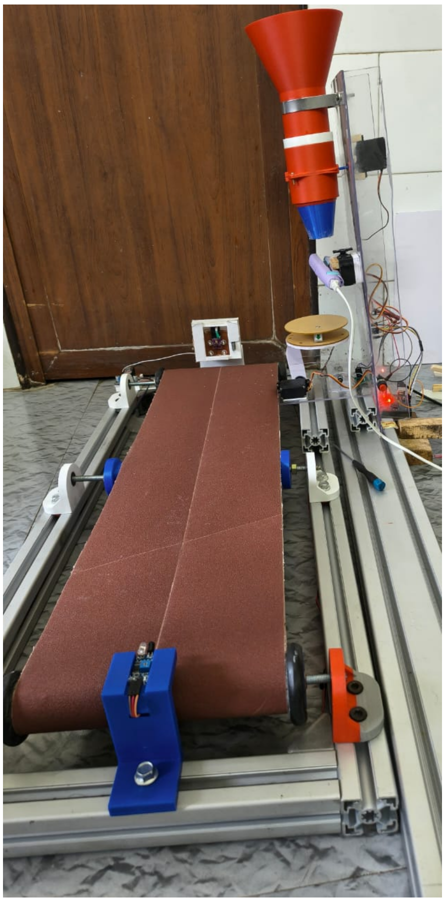

Figure 4.1: Photograph of Assembled Conveyor System

platform. Once detected, the stepper motor is activated to move the conveyor.

At the opposite end of the conveyor, a limit switch detects the packet's arrival, triggering the system to halt the conveyor motion. This start-stop mechanism ensures precise packet positioning and prevents packets from falling off the conveyor end.

4.4.1 IR Sensor Operation

- Activates conveyor when packet is detected.
- Ensures conveyor only runs when necessary, optimizing energy consumption.

4.4.2 Limit Switch Function

- Stops conveyor upon packet arrival at endpoint.

- Prevents unnecessary conveyor cycling.

Figure 4.2: Placement of IR Sensor and Limit Switch on Conveyor

4.5 Arduino-Based Control Logic

The conveyor system's control is integrated into the central Arduino Mega unit governing the entire prototype. Input from the IR sensor and limit switch is read through digital pins, with corresponding outputs sent to the stepper motor driver.

4.5.1 Arduino Code Snippet

```
#include <AccelStepper.h>
AccelStepper stepper(1, 2, 3); // Stepper control pins

void loop() {
    if (digitalRead(IR_sensor) == LOW) {
```

```
        stepper.setSpeed(200);
        stepper.runSpeed();
    }
    if (digitalRead(limit_switch) == LOW) {
        stepper.stop();
    }
}
```

4.6 Challenges Faced

The following challenges were encountered during the development of the conveyor system:

- Alignment precision was crucial to prevent packets from slipping off the belt.
- The IR sensor's sensitivity required fine-tuning to avoid false triggering.
- Belt tensioning had to be optimized for consistent motion without slippage.
- Integration with the tilting mechanism required careful positioning to avoid collision.

Through iterative testing and adjustments, these issues were systematically resolved to achieve reliable performance.

4.7 Summary

The conveyor system was successfully designed, fabricated, and integrated within the overall prototype to automate the packet transfer process. Its modular construction, responsive control logic, and sensor feedback integration ensure reliable operation, laying the foundation for future scaling and enhancements for industrial applications.

Chapter 5

System Integration with Sealing and Tilting Mechanisms

5.1 Introduction

A successful automation system relies not only on the functionality of its individual components but also on the seamless integration of these components into a synchronized workflow. In this project, the integration of the filling, weighing, sealing, and tilting mechanisms was carried out to ensure smooth operation from start to finish. This chapter elaborates on how the various subsystems were combined into a cohesive working prototype, highlighting the role of the sealing and tilting mechanisms in completing the automated packaging cycle.

5.2 Overview of Integrated System

The integration process focused on aligning the functionalities of the hopper, conveyor, sealing, and tilting mechanisms under the control of a centralized microcontroller (Arduino Mega). The objective was to automate the sequential steps of filling, weighing, sealing, and transferring TiO_2 packets without manual intervention, except for the initial placement of empty packets.

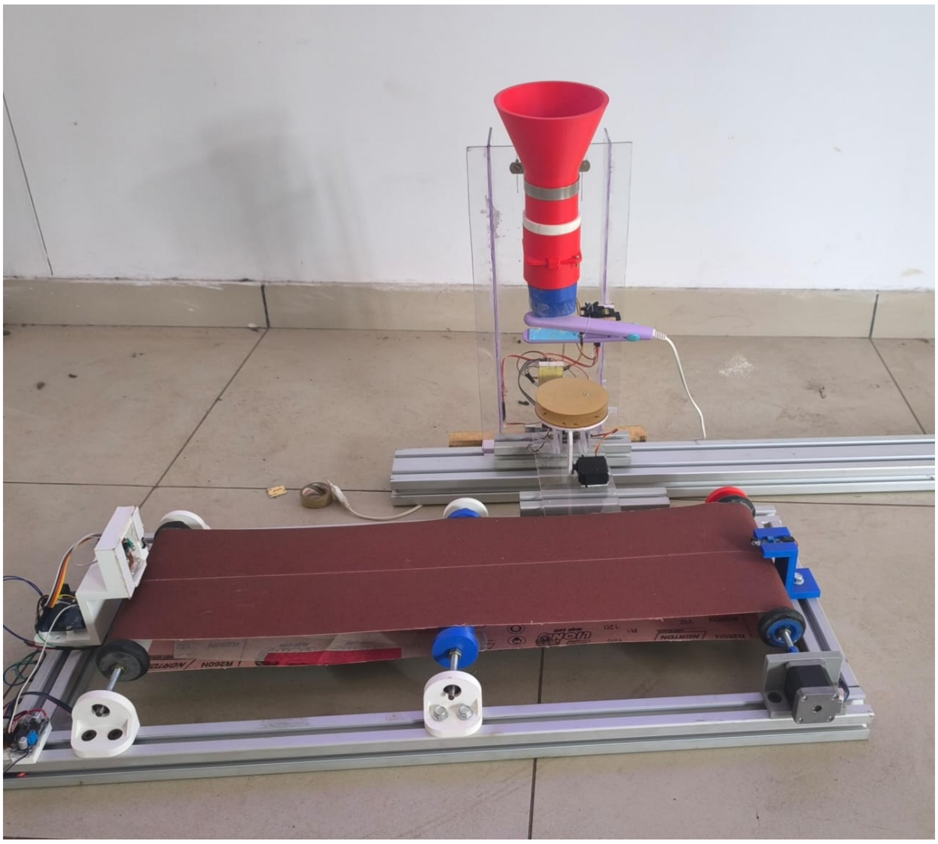

Figure 5.1: Photograph of Fully Integrated Prototype System

5.3 Sealing Mechanism Integration

The sealing mechanism plays a critical role in ensuring the packet's integrity and preventing tampering or leakage of the Titanium Dioxide powder. For this prototype, a heat sealer was implemented to simulate the sealing process, with plans for ultrasonic sealing proposed in the future.

5.3.1 Design and Actuation

The sealing unit was mounted in alignment with the load cell platform, ensuring that packets remain properly positioned during sealing. The opening and closing of the sealing jaws were controlled using a high-torque servo motor, allowing for precise timing and pressure application.

During operation, after the desired weight is reached and the butterfly valve closes, the Arduino triggers the servo to actuate the sealing unit. The servo rotates from its default open position (180 degree) to the closed sealing position (0 degree), holding for approximately 5 seconds to ensure a complete seal before returning to its original position.

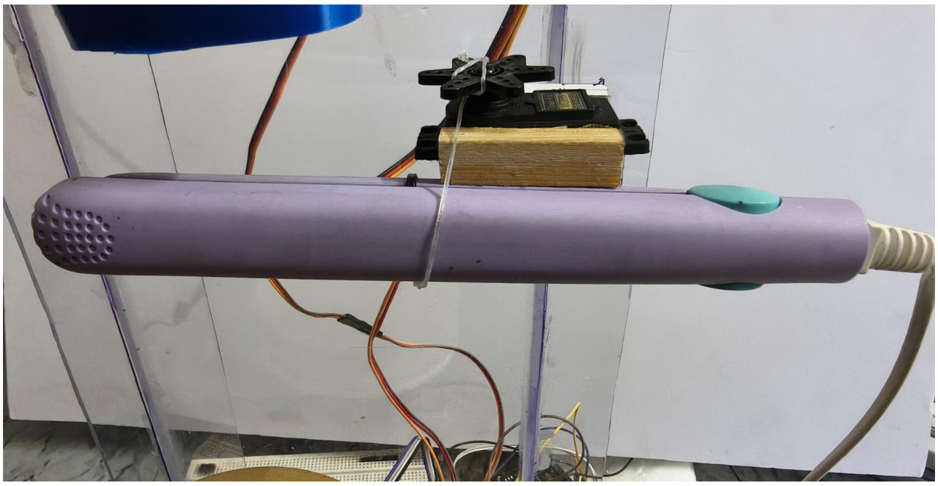

Figure 5.2: Photograph of Integrated Sealing Unit

5.4 Tilting Mechanism Integration

Following the sealing process, the packet must be safely and efficiently transferred onto the conveyor for further handling. This was achieved through a tilting platform mechanism integrated with the weighing system.

5.4.1 Design and Actuation

The weighing platform, which houses the load cell, was designed with a tilting feature controlled by another high-torque servo motor. Upon completion of sealing, the Arduino sends a signal to the servo, causing the platform to tilt from 0 degree to 45 degree, allowing the sealed packet to slide onto the conveyor belt below. Once the packet is transferred, the platform resets to its default horizontal position, ready to receive the next empty packet.

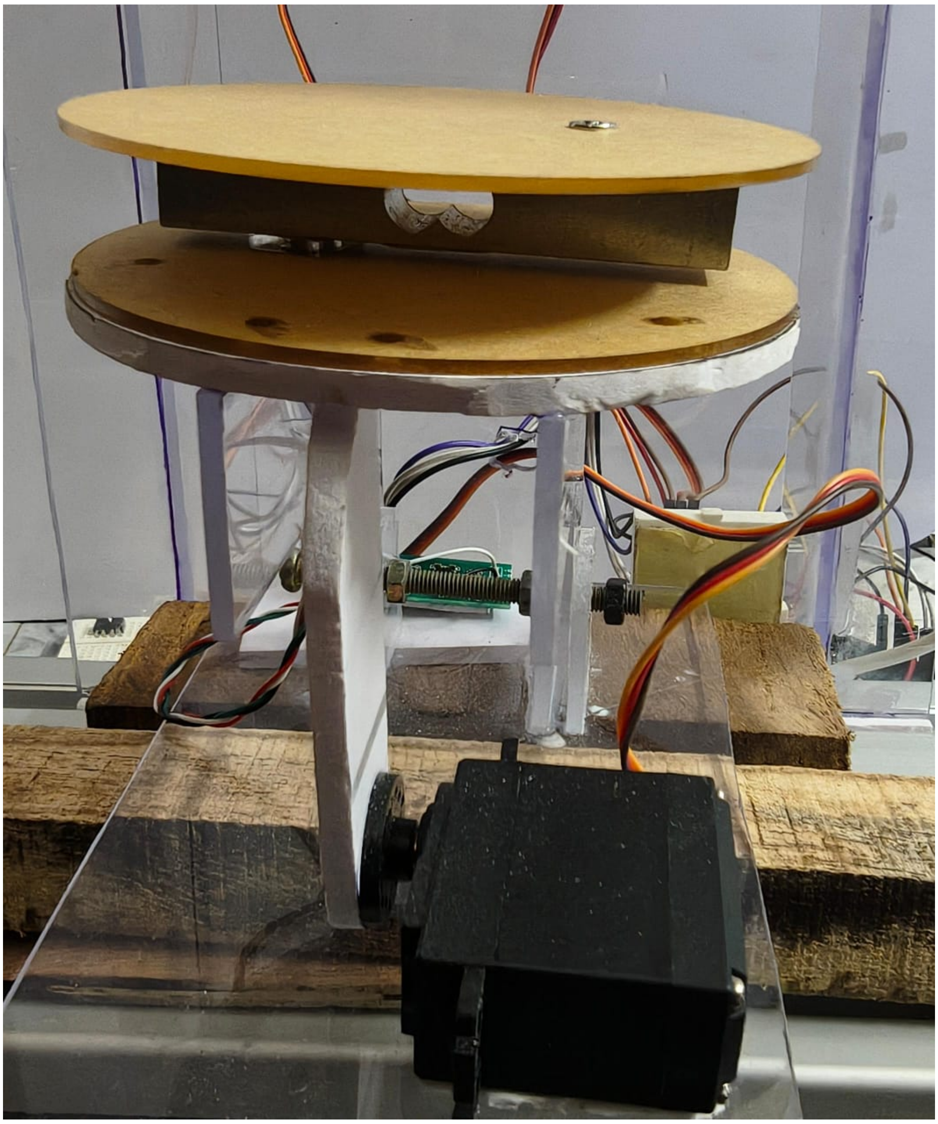

Figure 5.3: Photograph of Integrated Tilting Mechanism

5.5 Synchronization and Control Logic

The control system was programmed to ensure proper sequencing and timing of all mechanisms:

1. Packet placed on platform.
2. Filling begins until desired weight is reached.

3. Valve closes after target weight is detected.
4. Sealing unit actuates after a brief delay.
5. Sealing completes and platform tilts to release packet.
6. Conveyor transports the packet to the endpoint.

This synchronized flow ensures smooth operation without manual intervention beyond initial placement.

5.5.1 Arduino Control Logic (Sealing and Tilting)

```
Servo sealingServo, tiltServo;
sealingServo.attach(10);
tiltServo.attach(11);

if (weight >= 90) {
    delay(2000);
    sealingServo.write(0);  // Close for sealing
    delay(5000);
    sealingServo.write(180);  // Release after sealing
    delay(2000);
    tiltServo.write(45);  // Tilt to release packet
    delay(2000);
    tiltServo.write(0);  // Reset position
}
```

5.6 Challenges in Integration

Several integration challenges were encountered during the assembly and testing phases:

- Aligning the sealing unit precisely with the packet position to ensure consistent sealing.
- Calibrating servo angles to avoid excessive pressure on the packet or incomplete tilting.
- Ensuring timing synchronization to prevent overlapping actions between mechanisms.
- Structurally supporting the additional weight and motion of sealing and tilting components without disturbing load cell readings.

These challenges were systematically addressed through iterative testing, mechanical adjustments, and refinements in the control code.

5.7 Summary

The successful integration of the sealing and tilting mechanisms with the hopper and conveyor systems marked the completion of the automated packaging prototype. Each subsystem functions in coordination under centralized control, providing a reliable demonstration of automated filling, sealing, and transferring of TiO_2 packets. This integrated system lays the groundwork for future scalability and refinement for industrial applications.

Chapter 6

Overall System Working Methodology

6.1 Introduction

The successful development of this prototype lies not only in the design of individual components but in how effectively these subsystems were integrated and synchronized to achieve a smooth, automated process. This chapter explains the overall working methodology of the developed prototype, detailing how each component interacts within the sequence of operations, from the initial loading of the packet to its final transfer via the conveyor system.

6.2 Overview of Process Flow

The overall system is designed to replicate, on a prototype scale, the industrial process of automated weighing, filling, sealing, and conveying of TiO_2 packets. The following steps summarize the methodology:

1. An empty packet is manually placed on the load cell platform.
2. Upon initiation, the hopper's butterfly valve opens to allow powder flow.
3. The load cell continuously measures the packet's weight.

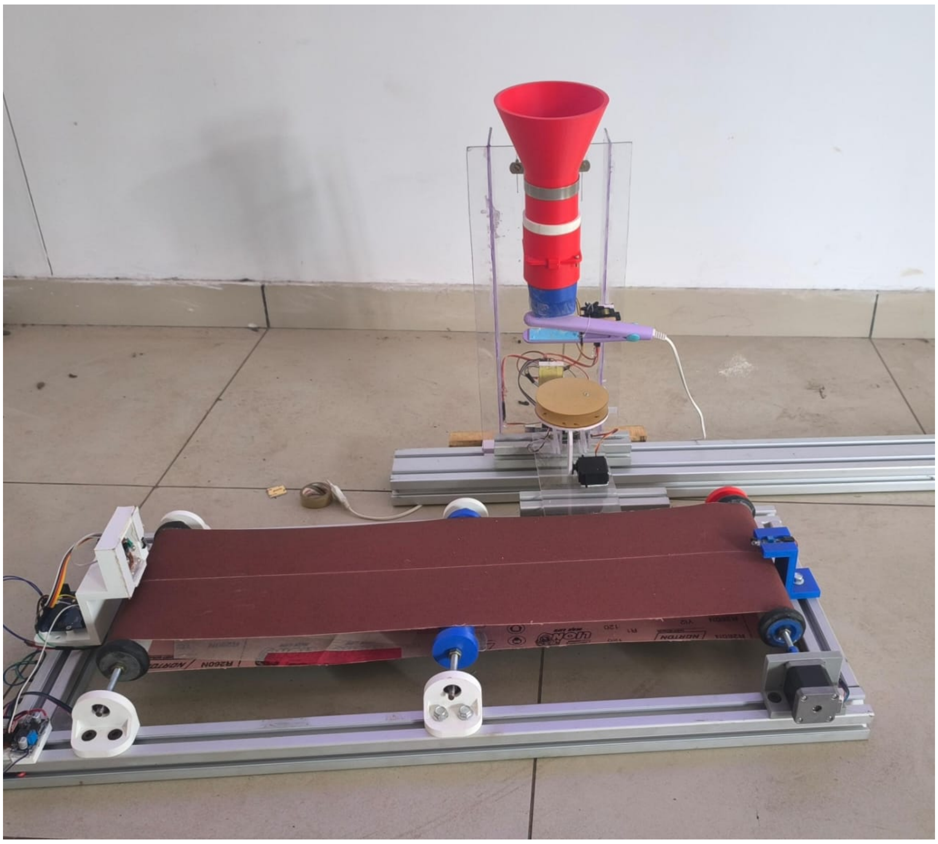

Figure 6.1: Photograph of Overall Assembled Prototype System

4. Once the target weight is reached, the valve closes.

5. After a brief delay, the sealing mechanism activates.

6. Following sealing, the platform tilts, releasing the packet onto the conveyor.

7. The conveyor transports the packet until the limit switch stops the motion.

6.3 Detailed Description of Working Methodology

6.3.1 Packet Placement and Weighing

The process begins with the manual placement of an empty packet onto the weighing platform, which incorporates a calibrated load cell. Once placed, the system is ready to initiate the filling sequence.

6.3.2 Filling Process

Upon starting, the Arduino Mega sends a control signal to open the servo-actuated butterfly valve. The Titanium Dioxide powder flows through the hopper and funnel, guided precisely onto the packet. The load cell continuously monitors the accumulating weight via the HX711 amplifier. When the measured weight reaches approximately 90-100 grams, the Arduino instructs the servo to close the valve, halting the powder flow.

6.3.3 Sealing Operation

With the packet filled, the sealing mechanism is activated after a short programmed delay to ensure stability. The servo-controlled heat sealer clamps the packet's opening, applies heat for a defined duration (typically 5 seconds), and then releases it. This ensures a secure seal, preventing powder leakage or tampering.

6.3.4 Tilting Mechanism

After sealing, the platform tilts to transfer the packet onto the conveyor. The servo motor actuates the tilting from 0 degree to 45 degree, allowing gravity to slide the packet gently onto the moving belt below. The platform then returns to its horizontal position, ready for the next cycle.

6.3.5 Conveyor Transfer

The IR sensor at the start of the conveyor detects the falling packet and signals the stepper motor to begin moving the belt. The conveyor continues to operate until the packet reaches the limit switch at the end, which then stops the motor, completing the transfer phase. This controlled movement ensures precise handling and avoids packet displacement.

6.4 Sequence Control and Timing

The system's entire operation is governed by a structured control logic programmed into the Arduino Mega. The process flow ensures no overlap occurs between subsystems and that each step is executed only after the completion of the previous one. This sequential logic avoids operational conflicts and maintains the integrity of the process.

6.4.1 Simplified Process Flow (Pseudocode)

```
Start Process
-> Open Valve -> Start Filling -> Monitor Weight
IF Weight >= 90g THEN
    Close Valve
    Delay
    Activate Sealing
    Delay
    Tilt Platform
    Delay
    Start Conveyor
    Stop Conveyor at Limit Switch
END IF
```

6.4.2 Arduino Control Code Implementation

The following Arduino code was developed and implemented to automate the sequential operations of the filling, sealing, tilting, and weighing mechanisms. The code utilizes the HX711 library for load cell communication and Servo library for controlling the servo motors associated with the valve, sealing mechanism, and tilting platform.

Listing 6.1: Arduino Control Code for System Operation

```
#include <Servo.h> // Servo library
#include "HX711.h" // Load cell library

// Pin Definitions
#define DOUT 3
#define CLK 2
#define buttonPin 7
#define valveServoPin 5
#define sealServoPin 6
#define tiltServoPin 9

// Servo Objects
Servo valveServo;
Servo sealServo;
Servo tiltServo;

// HX711 Object
HX711 scale;

// State Flags
bool isFilling = false;
bool isPostValveDelay = false;
bool isSealing = false;
bool isPostSealDelay = false;
bool isTilting = false;

// Timing Variables
unsigned long postValveDelayStart = 0;
unsigned long sealStartTime = 0;
unsigned long postSealDelayStart = 0;
unsigned long tiltStartTime = 0;
```

```
// Constants
const float targetWeight = 100.0; // grams
const int tiltDefaultPos = 45;
const int tiltTiltedPos = 0;
const int valveOpenPos = 70;
const int valveClosedPos = 0;
const int sealDownPos = 0;
const int sealIdlePos = 180;
const unsigned long postValveDelay = 1000;
const unsigned long postSealDelay = 2000;

// Calibration Factor
float calibration_factor = -103;

void setup() {
  Serial.begin(9600);
  scale.begin(DOUT, CLK);
  scale.set_scale(calibration_factor);
  scale.tare();

  valveServo.attach(valveServoPin, 500, 2400);
  sealServo.attach(sealServoPin, 500, 2400);
  tiltServo.attach(tiltServoPin, 500, 2400);

  valveServo.write(valveClosedPos);
  sealServo.write(sealIdlePos);
  tiltServo.write(tiltDefaultPos);

  pinMode(buttonPin, INPUT_PULLUP);

  Serial.println("System Ready. Press button to start.");
}

void loop() {
```

```
// Start cycle on button press
if (digitalRead(buttonPin) == LOW && !isFilling && !isPostValveDelay &&
!isSealing && !isPostSealDelay && !isTilting) {
  isFilling = true;
  Serial.println("Filling Started...");
  valveServo.write(valveOpenPos);
  delay(200);
}
// Filling State
if (isFilling) {
  float weight = scale.get_units(2);
  Serial.print("Weight: ");
  Serial.print(weight, 2);
  Serial.println(" g");

  if (weight >= targetWeight) {
    valveServo.write(valveClosedPos);
    Serial.println("Target Weight Reached. Valve Closed.");
    delay(200);

    isFilling = false;
    isPostValveDelay = true;
    postValveDelayStart = millis();
    Serial.println("Waiting before sealing...");
  }
}

// Delay after valve closes
if (isPostValveDelay) {
  if (millis() - postValveDelayStart >= postValveDelay) {
    isPostValveDelay = false;
    isSealing = true;
    sealStartTime = millis();
    sealServo.write(sealDownPos);
```

```
      Serial.println("Sealing Started...");
    }
  }
  // Sealing State
  if (isSealing) {
    if (millis() - sealStartTime >= 2000) {
      sealServo.write(sealIdlePos);
      isSealing = false;
      isPostSealDelay = true;
      postSealDelayStart = millis();
      Serial.println("Sealing Done. Waiting before tilting...");
    }
  }

  // Delay after sealing
  if (isPostSealDelay) {
    if (millis() - postSealDelayStart >= postSealDelay) {
      isPostSealDelay = false;
      isTilting = true;
      tiltStartTime = millis();
      tiltServo.write(tiltTiltedPos);
      Serial.println("Tilting Started...");
    }
  }
  // Tilting State
  if (isTilting) {
    if (millis() - tiltStartTime >= 3000) {
      tiltServo.write(tiltDefaultPos);
      isTilting = false;
      scale.tare();
      Serial.println("Tilting Done. Ready for next cycle.");
    }
  }
}
```

6.5 Integration Benefits

The integrated operation of all subsystems demonstrates the following advantages:

- Reduces human involvement in potentially hazardous operations.
- Ensures consistent filling and sealing quality.
- Minimizes material wastage through accurate weighing and controlled dispensing.
- Streamlines the packet transfer process for improved workflow efficiency.

6.6 Summary

This chapter presented a detailed description of the prototype's overall working methodology. The smooth interaction between the hopper, weighing platform, sealing unit, tilting mechanism, and conveyor system successfully demonstrates a fully automated process flow. This serves as a proof-of-concept for scaling up to industrial requirements, where larger volumes and stricter controls would apply.

Chapter 7

Testing and Results

7.1 Introduction

Testing plays a crucial role in validating the performance and functionality of any engineering system before its real-world implementation. For this project, which involves the design and development of an automated prototype for the weighing, filling, sealing, and conveying of Titanium Dioxide (TiO_2) packets, the testing phase was carried out meticulously to verify the reliability of both individual subsystems and the integrated system. The goal of this phase was to ensure that the developed prototype performs its intended functions accurately, efficiently, and consistently.

This chapter discusses the testing procedures undertaken, the methodology followed, observations made during the process, and the analysis of the prototype's performance under practical conditions.

7.2 Testing Objectives

The primary objectives of testing the developed prototype were as follows:

- To verify the accuracy of the load cell in detecting and measuring the weight of the packets.

- To ensure the controlled actuation of the butterfly valve during the filling operation.
- To confirm the sealing mechanism's ability to secure the packets properly.
- To validate the tilting mechanism's performance in transferring the packets to the conveyor.
- To assess the synchronization and smooth functioning of the overall system.

7.3 Testing Methodology

The testing process was methodically divided into two stages: **individual subsystem testing** and **integrated system testing**. This approach ensured that each component functioned correctly on its own before verifying its interaction with the complete system.

7.3.1 Individual Subsystem Testing

Load Cell Calibration and Testing

The load cell, paired with the HX711 amplifier module, was calibrated using standard reference weights. This calibration ensured the accuracy of weight readings throughout the operation. The calibration factor was adjusted iteratively until the measured weights consistently matched the known values within an acceptable margin of error.

Butterfly Valve Testing

The servo-controlled butterfly valve was tested for repeatability and precision. Its ability to open and close smoothly, allowing controlled material flow without clogging or leakage, was verified through multiple trials.

Sealing Mechanism Testing

The heat sealing mechanism was evaluated by sealing sample packets under varied conditions. Observations were focused on achieving uniform sealing strength and verifying the sealing duration controlled by the servo mechanism.

Tilting Platform Testing

The tilting platform's performance was assessed under load conditions to confirm that it could reliably tilt to transfer filled packets onto the conveyor and return to its default position without causing packet spillage or instability.

Conveyor Testing

The conveyor system was tested independently to confirm its response to IR sensor inputs and its ability to transport packets accurately to the endpoint where a limit switch would halt its motion.

7.3.2 Integrated System Testing

Once individual components performed satisfactorily, the complete system was assembled, and integrated testing was conducted. The testing sequence mirrored the actual operation of the system:

1. Placement of an empty packet on the weighing platform.
2. Activation of the filling mechanism via the servo-controlled valve.
3. Monitoring of weight accumulation through the load cell.
4. Automatic sealing once the target weight was achieved.
5. Tilting the platform to transfer the packet onto the conveyor.

6. Conveyor transport and halting upon reaching the limit switch.

Multiple test cycles were carried out to evaluate consistency, repeatability, and synchronization between components.

7.4 Observations and Results from Testing

During testing, several observations were made, leading to minor adjustments for optimal performance. These are summarized below:

- **Weighing System:** The load cell delivered consistent results, with weight readings accurate within ±5 grams of the target value. Regular taring before each cycle ensured precision.
- **Filling Mechanism:** The servo-actuated butterfly valve functioned reliably, halting material flow precisely upon reaching the set weight threshold.
- **Sealing Unit:** The sealing process consistently produced sealed packets suitable for prototype demonstration. However, alignment was found to be critical for uniform sealing.
- **Tilting Platform:** The tilting mechanism operated smoothly, transferring packets effectively without disturbing the contents.
- **Conveyor:** The IR sensor accurately detected packets, and the stepper motor-driven conveyor stopped correctly at the limit switch, ensuring controlled movement.

No major system failures occurred during integrated testing, and all components worked in harmony under the programmed logic.

7.5 Discussion

The testing phase demonstrated that the prototype successfully fulfills its intended functions. While designed for handling 100 grams, the prototype effectively mirrors the working principles required for industrial-scale automation. Key lessons learned during testing include the importance of precise calibration, careful mechanical alignment, and synchronization of sequential operations.

Minor issues such as timing adjustments for servo responses and optimal sealing durations were resolved through iterative testing, leading to improved system reliability. These findings highlight areas for future refinement when scaling the system to accommodate heavier packets and more robust sealing methods like ultrasonic sealing.

7.6 Summary

In summary, the testing process validated the performance of the developed prototype. Each subsystem met its functional requirements, and the integrated system demonstrated smooth, automated operation from weighing to final packet transfer. The results confirm that this prototype serves as a practical foundation for future scaling and refinement towards meeting the industrial standards expected by Travancore Titanium Products Limited.

Chapter 8

Proposed Future Developments

8.1 Introduction

While the current prototype successfully demonstrates the automation of Titanium Dioxide (TiO_2) weighing, filling, sealing, and conveying on a small scale, several proposed enhancements remain unrealized within this stage of the project. These proposals, identified through feedback from Travancore Titanium Products Limited (TTPL) and through critical assessment of industrial best practices, are intended to strengthen the system's scalability, safety, and commercial viability in future iterations.

This chapter outlines these proposals, detailing how they can address existing limitations and improve upon the current prototype.

8.2 Limitations of the Current Prototype

The existing prototype was intentionally designed to function within the constraints of a small-scale demonstration. Its primary limitations include:

- Manual placement of empty packets.
- Limited to 100 grams, whereas industrial applications require 25 kg handling.

- Use of basic heat sealing, which lacks tamper-proof integrity compared to ultrasonic sealing.
- Absence of dust mitigation systems.
- Lack of data logging or digital monitoring interfaces.

Recognizing these gaps allows for a clear roadmap towards industrial readiness.

8.3 Proposed Enhancements

8.3.1 Automated Packet Feeding Mechanism

In future versions, the manual placement of empty packets can be eliminated through the addition of an automated packet feeder. This enhancement will not only improve efficiency but also reduce human exposure to TiO_2, aligning with occupational safety standards.

8.3.2 Scaling Up to 25 kg Capacity

The system must be structurally and mechanically reinforced to handle heavier packets (up to 25 kg), as required by TTPL's operational specifications. This would involve upgrading the load cell capacity, conveyor motor specifications, and ensuring robust mechanical supports.

8.3.3 Ultrasonic Sealing for Tamper-Proofing

One of the most significant proposed upgrades is the replacement of the basic heat sealing mechanism with an ultrasonic sealing system. Ultrasonic sealing offers superior consistency, is tamper-proof, and is already widely accepted in the packaging industry for its reliability and clean sealing capabilities.This method would ensure airtight seals without thermal damage to the packaging material, further enhancing product quality and safety.

Figure 8.1: Representative Image of Industrial Ultrasonic Sealing Equipment

Source: https://fair888.en.made-in-china.com/product/DJhUgNwvgnpL/China-Cement-Tile-Adhesive-Automatic-Ultrasonic-Powder-Filling-Machine-Dry-Mortar-Packing-Machine-Industrial-Machinery.htmlMade-in-China

8.3.4 Enclosed System with Dust Management

To further protect workers and reduce environmental contamination, the future system can be fully enclosed with integrated dust collection mechanisms. This enclosure would contain any airborne TiO_2 during the filling process, complying with industrial hygiene standards.

8.3.5 Digital Interface and Data Logging

Integrating a Human-Machine Interface (HMI) and data logging features will allow for process monitoring, error tracking, and production analysis. Such systems are standard in modern manufacturing and would enable TTPL to monitor output, error rates, and maintenance needs efficiently.

8.3.6 Enhanced Safety Features

Future iterations may include:

- Emergency stop systems.
- Automated safety interlocks.
- Load imbalance detection.
- Overfill prevention safeguards.

These features would further enhance operational safety, particularly at industrial scales.

8.4 Roadmap for Implementation

A structured development roadmap is suggested for transitioning from prototype to industry-ready system:

1. Validate scaled-up designs through simulations and small-batch testing.
2. Integrate ultrasonic sealing as a core feature.
3. Develop automated packet handling mechanisms.
4. Implement dust management solutions.
5. Incorporate digital monitoring and data logging.
6. Conduct industrial trials in partnership with TTPL for final validation.

8.5 Summary

While this project successfully meets the objectives of demonstrating an automated TiO_2 handling system at a prototype level, its future lies in expanding capacity, improving safety, and aligning with industrial expectations. The proposed enhancements, particularly the adoption of ultrasonic sealing and full automation, will transform this prototype into a scalable, industry-compliant solution capable of delivering efficiency, consistency, and safety at TTPL's production scale.

Chapter 9

Conclusion

9.1 Introduction

This project focused on the design and development of a budget-friendly, prototype-scale automated system for the weighing, filling, sealing, and conveying of Titanium Dioxide (TiO_2) packets at Travancore Titanium Products Limited (TTPL). The initiative aimed to address the challenges of manual handling, improve operational efficiency, and enhance workplace safety by reducing human exposure to TiO_2.

Through careful planning, iterative design, and systematic testing, the project successfully developed a functional prototype capable of demonstrating the core principles of automation required for this industrial application.

9.2 Summary of Work Done

The project was systematically divided into multiple focused areas, with each subsystem developed and tested individually before integrating into the final assembly:

- A load cell-based weighing mechanism was developed to ensure precise material measurement.

- A servo-controlled butterfly valve managed the controlled dispensing of TiO_2 powder.
- A basic heat sealing mechanism simulated the industrial sealing process to prevent material leakage.
- A tilting platform was designed to automate packet transfer from the weighing station to the conveyor.
- A conveyor system, integrated with sensors, enabled the smooth transportation of packets through the process.

Each subsystem was successfully controlled through an Arduino Mega microcontroller, providing centralized automation of the sequential operations.

9.3 Achievements

The project successfully met its objectives within the prototype's defined scope:

- Demonstrated the feasibility of automating TiO_2 weighing and packaging processes.
- Achieved accurate weighing and reliable flow control through servo-actuated mechanisms.
- Ensured consistent sealing suitable for prototype validation.
- Integrated sequential automation of filling, sealing, tilting, and conveying processes.
- Validated system functionality through repeated testing with positive results.

These achievements confirm the practicality of further developing the system towards an industrial-scale solution.

9.4 Challenges Overcome

Throughout the project, several technical and operational challenges were encountered:

- Fine TiO_2 powder presented difficulties in flow control, requiring iterative hopper and valve adjustments.
- Synchronizing servo mechanisms for sealing and tilting demanded careful calibration and programming.
- Load cell calibration required precision to maintain measurement accuracy.
- Mechanical alignment issues, particularly in the sealing and tilting mechanisms, were addressed through design refinements.

Overcoming these challenges provided valuable insights into the complexities of developing automated material handling systems.

9.5 Significance of the Project

The project provides a strong foundation for TTPL to transition towards automated packaging processes. Although scaled for 1kg prototypes, the system principles are directly applicable to the company's 25kg production requirements. The reduction in manual handling of TiO_2 will significantly minimize occupational health risks and improve overall operational efficiency.

Moreover, this prototype serves as a reference model for integrating similar automation solutions in industries handling fine powders or granular materials.

9.6 Future Scope

Building upon this successful prototype, future work should focus on:

- Scaling the system to handle higher weights (25kg).
- Implementing ultrasonic sealing for enhanced tamper-proofing.
- Incorporating automated bag feeding mechanisms.
- Adding data logging and digital interfaces for real-time monitoring.
- Integrating advanced safety features and dust management systems.

These developments will align the system with modern industrial standards and strengthen TTPL's operational safety and efficiency.

9.7 Final Remarks

The successful completion of this project validates the approach and confirms the potential for fully automating the TiO_2 weighing and packaging process at TTPL. The prototype serves not only as a proof-of-concept but also as a practical stepping stone towards a safer, more efficient, and technologically advanced production environment.

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